

Expanding SEPSES Cybersecurity Knowledge Graphs (CSKG)

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Abstract. Modern cybersecurity faces the challenge of data fragmentation, where information concerning malware behavior (MAEC), threat sources (MISP Threat Intelligence), adversary techniques (MITRE ATT&CK), and mitigation standards (NIST SP 800-53 Security Controls) resides in isolated data silos. The objective of this research is to overcome this fragmentation through the construction of an Integrated Knowledge Graph (KG). We adopted a triple-based approach with a tailored ontology schema. The integration process involved RML (R2RML/RML) mapping to convert heterogeneous source data (JSON/TTL) into RDF triples, followed by crucial semantic linking using custom properties (`sec:isMitigatedBy` and `ti:hasObservedIndicator`). The resulting Knowledge Graph was loaded into a SPARQL Endpoint (Apache Jena Fuseki) and evaluated. Results demonstrated full success in traversing the threat path (MISP → MAEC → ATT&CK → NIST), enabling complex queries such as identifying the recommended NIST Control (AC-3: Access Enforcement) to mitigate a specific Malware Action (`WriteFile`) observed from an external threat feed (ELLIO Community Feed). This integrated KG successfully provides a holistic framework for rapid risk analysis and mitigation decision-making.

Keywords: Knowledge Graph, RDF, RML, SPARQL, MAEC, MISP, MITRE ATT&CK, NIST SP 800-53, Semantic Data Integration, Threat Intelligence

1 Introduction

Cybersecurity information is produced across many heterogeneous sources, including vulnerability databases, malware reports, attack pattern catalogs, and threat intelligence feeds. While these resources such as CVE, CWE, CVSS, CPE, and CAPEC provide essential insights, they are published in different formats and lack semantic interoperability, making it difficult for security analysts to correlate and reason over them efficiently.

Cybersecurity Knowledge Graphs (CSKGs) offer a promising solution by integrating diverse security datasets into a unified, machine-interpretable representation. The SEPSES CSKG demonstrates this potential by linking major

industry standards and maintaining them through an automated ETL workflow [1]. Similar efforts, such as ICS-SEC KG, further validate the effectiveness of structured, interlinked cybersecurity knowledge for specialized domains like industrial control systems [2]. Recent surveys highlight that such knowledge graphs significantly improve threat analysis, knowledge reuse, and automated security reasoning [3], [4].

However, the current SEPSES CSKG still excludes several important structured sources that could greatly enhance analytical depth. Prior work emphasizes the need to incorporate additional formats such as STIX/TAXII for threat sharing [5], MAEC for malware behavior modeling, MISP for real-world incident feeds, and NIST frameworks for security and privacy controls. Integrating these sources enables richer cross-model analysis of adversary behavior, vulnerabilities, and mitigation strategies.

In response, this project expands the SEPSES CSKG by integrating four additional structured cybersecurity resources: **MAEC**, **MITRE ATT&CK**, **NIST Privacy Controls**, and **MISP**. We developed custom ontologies, created RML mappings to transform raw data into RDF, and linked the resulting entities with existing SEPSES components using shared identifiers such as CVE and ATT&CK IDs. This extended graph supports more comprehensive analysis, including threat traceability, malware behavior enrichment, and linkage between threat events and security controls. To demonstrate its utility, we present three real-world use cases based on the integrated KG.

2 Related Work

Cybersecurity Knowledge Graphs (CSKGs) have emerged as a powerful approach for integrating heterogeneous cybersecurity information into unified, machine-interpretable structures that support threat analysis, vulnerability assessment, and automated reasoning. Prior work such as the SEPSES CSKG provides a foundational framework by semantically integrating widely used standards—including CVE, CWE, CVSS, CAPEC, and CPE—and maintaining them through a continuous ETL pipeline [1].

Similar extensions have been explored in domain-specific contexts, most notably the ICS-SEC KG, which aggregates industrial control system advisories, vulnerabilities, and attack techniques to support situational awareness in operational technology environments [2]. Beyond these efforts, several studies highlight the need for broader semantic integration across diverse CTI formats. Existing ontologies such as UCO unify major standards into a common conceptual model, while other works investigate the representation of threat intelligence using formats like STIX/TAXII [5], malware behavior through MAEC, and community-driven incident sharing platforms such as MISP.

Surveys of cybersecurity KGs consistently underline their impact in enabling knowledge reuse, adversary behavior modeling, and scalable security automation [3], [4]. However, despite these advances, current CSKGs still lack unified integration of threat events (MISP), malware actions (MAEC), adversary techniques (ATT&CK), and security/privacy controls (NIST) within a single interoperable graph. This gap motivates our work to expand the SEPSES CSKG with additional structured sources, thereby

enabling richer cross-model reasoning and more comprehensive threat intelligence analysis.

3 Proposed Approach

3.1 Overall Framework

The construction of the extended CSKG follows a five-stage automated workflow that transforms four structured cybersecurity sources into an integrated and queryable knowledge graph. The complete pipeline is illustrated in Fig. X.

1. **Data Acquisition and Pre-processing**

All selected sources are collected in JSON format. The structured input datasets used in this work were obtained from publicly available machine-readable repositories, including ATT&CK [6], MAEC [7], MISP [8], NIST SP 800-53 [9]. Light preprocessing is applied when necessary to clean or align fields, ensuring that the JSON structures are consistent and suitable for RML transformation.

2. **Ontology Development**

Four domain-specific ontologies (MAEC, MISP, ATT&CK, and NIST) are designed in Turtle format using a bottom-up approach. Each ontology defines the core classes, object properties, and datatype properties required to model the semantics of its corresponding source.

3. **RML Mapping and RDF Generation**

For each source, an RML mapping file is created and executed using RMLMapper.jar to translate normalized JSON data into RDF triples aligned with the ontologies. This step produces four separate Turtle files, one for each resource.

4. **Integration (Merging and Linking)**

The individual RDF outputs are merged into a unified graph (merged.ttl) and enriched with additional linking triples (linking.ttl) to produce the complete integrated knowledge graph (full-kg.ttl). This step aligns related entities across sources using shared identifiers and ontology properties.

5. **Validation and Deployment**

Optional SHACL validation is applied to verify structural and semantic correctness. The final knowledge graph is then deployed to Virtuoso as a named graph and exposed through a SPARQL endpoint for querying and analysis.

Fig. 1 provides an overview of the full pipeline, covering all stages from source ingestion to SPARQL deployment.

3.2 Ontology Development

To support semantic integration of the four structured cybersecurity sources (MAEC, MISP, MITRE ATT&CK, and NIST SP 800-53) we developed four domain ontologies in Turtle (TTL) format. Each ontology was constructed manually using a bottom-up modeling approach based directly on the underlying JSON structure of each dataset. The ontologies define the core classes, object properties, and datatype properties required for RML mapping and cross-source linkage. Representative excerpts from the actual ontology files are included below.

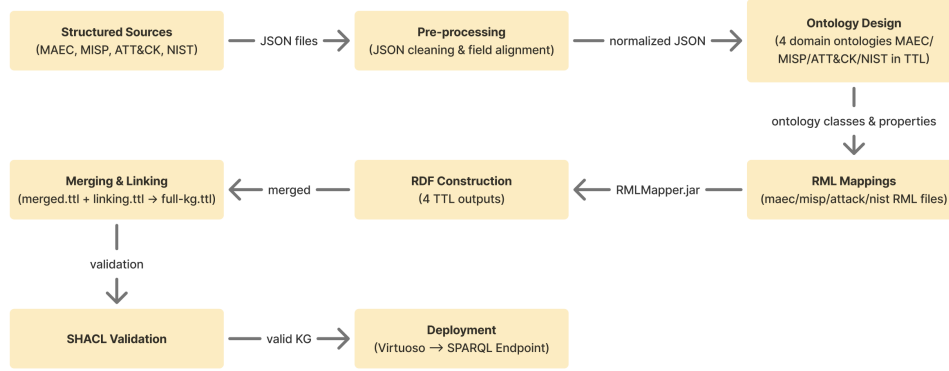


Fig. 1: Overall framework of the CSKG construction pipeline

1. ATT&CK Ontology

The ATT&CK ontology models adversary behavior, including threat groups, tactics, and techniques. It defines classes such as `attack:Technique` and `attack:AdversaryGroup`, along with properties linking groups to the techniques they employ. A snippet taken directly from the uploaded `attack-ontology.ttl` in Listing 1.10.

Listing 1.1: ATT&CK Ontology in TTL Format

```

@prefix attack: <http://w3id.org/sepses/vocab/ref/attack#> .

attack:AdversaryGroup rdf:type owl:Class ;
  rdfs:label "AdversaryGroup" .

attack:Technique rdf:type owl:Class ;
  rdfs:label "Technique" .

attack:usesTechnique rdf:type owl:ObjectProperty ;
  rdfs:domain attack:AdversaryGroup ;
  rdfs:range attack:Technique .

attack:techniqueID rdf:type owl:DatatypeProperty ;
  rdfs:range xsd:string .
  
```

This structure allows adversary groups to be linked to the techniques they employ, enabling later alignment with MAEC actions or NIST controls.

2. MAEC Ontology

The MAEC ontology represents malware behavior at the action level. It models atomic actions, API calls, and execution parameters, enabling fine-grained representation of malware behavior. The following Listing 1.11 excerpt from `maec-ontology.ttl` illustrates its core structure:

Listing 1.2: MAEC Ontology in TTL Format

```

@prefix maec: <http://w3id.org/sepses/vocab/ref/maec#> .

maec:MalwareAction rdf:type owl:Class ;
  rdfs:label "Malware_Action" ;
  rdfs:comment "A_single_atomic_action_performed_by_malware." .

maec:APICall rdf:type owl:Class ;
  rdfs:label "API_Call" .

maec:hasAPICall rdf:type owl:ObjectProperty ;
  rdfs:domain maec:MalwareAction ;
  rdfs:range maec:APICall .

maec:actionName rdf:type owl:DatatypeProperty ;
  rdfs:range xsd:string .

```

These constructs directly reflect the MAEC schema and allow granular behavioral modeling that can be linked to ATT&CK techniques.

3. MISP Ontology

The MISP ontology captures OSINT feed metadata, events, and indicators. It includes alignment with external vocabularies such as Schema.org, Dublin Core, and SKOS to enhance interoperability. The following Listing 1.12 excerpt from `misp-ontology.ttl` illustrates its core structure:

Listing 1.3: MISP Ontology in TTL Format

```

@prefix misp: <http://w3id.org/sepses/vocab/ref/misp#> .
@prefix mispf: <http://w3id.org/sepses/vocab/ref/misp/feed/> .
@prefix schema: <http://schema.org/> .
@prefix dct: <http://purl.org/dc/terms/> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .

mispf:Feed rdf:type owl:Class ;
  rdfs:label "MISP_Feed" .

misp:Event rdf:type owl:Class ;
  rdfs:label "MISP_Event" .

misp:hasIndicator rdf:type owl:ObjectProperty ;
  rdfs:domain misp:Event ;
  rdfs:range misp:Attribute .

mispf:feedName rdf:type owl:DatatypeProperty ;
  rdfs:subPropertyOf schema:name .

```

This ontology supports mapping MISP feed metadata, events, and indicators into RDF and linking them to external identifiers such as CVEs.

4. NIST Ontology

The NIST ontology formalizes the structure of NIST SP 800-53 controls based on the OSCAL JSON catalog. It includes classes for controls, profiles, and metadata, along with object properties linking them. The following Listing 1.13 excerpt from `nist-ontology.ttl`:

Listing 1.4: NIST Ontology in TTL Format

```
@prefix nist: <http://w3id.org/sepses/vocab/ref/nist#> .

nist:Control rdf:type owl:Class ;
  rdfs:label "Control" .

nist:Profile rdf:type owl:Class ;
  rdfs:label "Profile" .

nist:Metadata rdf:type owl:Class ;
  rdfs:label "Metadata" .

nist:hasMetadata rdf:type owl:ObjectProperty ;
  rdfs:domain nist:Profile ;
  rdfs:range nist:Metadata .

nist:controlID rdf:type owl:DatatypeProperty ;
  rdfs:range xsd:string .
```

5. Availability of Full Ontologies

All ontology files (ATT&CK, MAEC, MISP, and NIST), together with the RML mappings, RDF outputs, and KG integration scripts, are publicly available in the project's GitHub repository: https://github.com/Aria1516/Kelompok-1_CSKG

3.3 RML Mapping Strategy

To transform the four cybersecurity sources from JSON into RDF, we designed four RML mapping files—one for each ontology: MAEC, MISP, ATT&CK, and NIST. Each mapping file was created in Turtle (TTL) following the RML specification, consisting of `rr LogicalSource`, `rr SubjectMap`, and multiple `rr PredicateObjectMap` definitions. These mappings define how JSON fields are converted into ontology-aligned RDF triples. Representative excerpts from each mapping file are included below.

1. ATT&CK Mapping

The ATT&CK mapping (`attack_mapping.rml.ttl`) extracts techniques and adversary groups from the ATT&CK STIX/JSON file. It constructs technique URIs using technique IDs and maps labels, descriptions, and relationships to the ontology properties. An excerpt from the mapping file: An excerpt from the mapping file in Listing 1.14.

Listing 1.5: ATT&CK RML Mapping Snippet

```
<#TechniqueMapping>
rml:logicalSource [
  rml:source "enterprise-attack.json";
  rml:referenceFormulation ql:JSONPath;
  rml:iterator "$.objects[?(@.type=='attack-pattern')]"
] ;

rr:subjectMap [
```

```

rr:template "http://w3id.org/sepses/resource/attack/{id}" ;
rr:class attack:Technique
] ;

rr:predicateObjectMap [
  rr:predicate attack:techniqueID ;
  rr:objectMap [ rml:reference "external_references[0].external_id"
  ]
] ;

```

This mapping ensures that each ATT&CK technique is represented as an RDF resource with its identifier, description, and reference links.

2. MAEC Mapping

The MAEC mapping (maec_mapping.rml.ttl) transforms malware-action JSON data into RDF based on the MAEC ontology. It maps each malware action, its API calls, and associated parameters. Example excerpt in Listing 1.15:

Listing 1.6: MAEC RML Mapping Snippet

```

<#MalwareActionMapping>
rml:logicalSource [
  rml:source "malware-action.json";
  rml:referenceFormulation ql:JSONPath;
  rml:iterator "$.actions[*]"
] ;

rr:subjectMap [
  rr:template "http://w3id.org/sepses/resource/maec/action/{id}" ;
  rr:class maec:MalwareAction
] ;

rr:predicateObjectMap [
  rr:predicate maec:actionName ;
  rr:objectMap [ rml:reference "name" ]
] ;

```

This structure allows granular MAEC malware actions to be semantically enriched and later linked to ATT&CK techniques.

3. MISP Mapping

The MISP mapping (misp_mapping.rml.ttl) handles feed metadata and event structures from defaults.json. It models feeds, categories, and attributes using the MISP ontology. Example snippet in Listing 1.16:

Listing 1.7: MISP RML Mapping Snippet

```

<#FeedMapping>
rml:logicalSource [
  rml:source "defaults.json";
  rml:referenceFormulation ql:JSONPath;
  rml:iterator "$.defaults[*]"
] ;

```

```

rr:subjectMap [
  rr:template "http://w3id.org/sepses/resource/misp/feed/{feed-name}" ;
  rr:class mispf:Feed
] ;

rr:predicateObjectMap [
  rr:predicate mispf:feedName ;
  rr:objectMap [ rml:reference "name" ]
] ;

```

The mapping correctly aligns feed metadata to the ontology, including subproperty alignment such as schema:name and dct:identifier.

4. NIST Mapping

The NIST mapping (nist_mapping.rml.ttl) extracts NIST SP800-53 controls from the OSCAL JSON catalog. It maps control identifiers, titles, and associated meta-data. The following Listing 1.8 excerpt from nist-mapping.ttl:

Listing 1.8: NIST RML Mapping Snippet

```

<#ControlMapping>
rml:logicalSource [
  rml:source "NIST_SP-800-53_rev5_catalog-min.json";
  rml:referenceFormulation ql:JSONPath;
  rml:iterator "$.catalog.controls[*]"
] ;

rr:subjectMap [
  rr:template "http://w3id.org/sepses/resource/nist/control/{id}" ;
  rr:class nist:Control
] ;

rr:predicateObjectMap [
  rr:predicate nist:controlID ;
  rr:objectMap [ rml:reference "id" ]
] ;

```

All four RML mappings were executed using RMLMapper.jar, generating one Turtle file per source:

1. attack_output.ttl
2. maec_output.ttl
3. misp_output.ttl
4. nist_output.ttl

These output files form the base RDF graphs that are merged and interlinked in the next stage. The complete RML mapping files are available in the GitHub repository.

3.4 Entity Linking Strategy

The integration of the four independently generated RDF graphs is achieved through a combination of dataset merging and cross-ontology linking. This step produces the final

unified knowledge graph (full-kg.ttl), enabling entities from MAEC, MISP, ATT&CK, and NIST to interoperate within a consistent semantic space.

1. Merging Process

The RDF outputs generated from the four RML mappings (maec.ttl, misp.ttl, attack.ttl, and nist.ttl) were first combined into a single intermediate file, merged.ttl. This file consolidates all entities while preserving their original namespaces and ontology structures.

2. Ontology-level Alignment

Cross-source alignment was introduced in the linking.ttl file. Rather than enforcing direct entity-to-entity mappings, the linking file focuses on vocabulary alignment, ensuring interoperability across datasets and with external schemas. Examples taken directly from the linking file include:

Listing 1.9: Cross-Source Vocabulary Alignment

```
mispf:feedName rdfs:subPropertyOf schema:name .
mispf:feedSlug rdfs:subPropertyOf dct:identifier .
mispf:tagName rdfs:subPropertyOf skos:prefLabel .
mispf:tagSlug rdfs:subPropertyOf skos:notation .
```

These alignments allow MISP metadata to integrate seamlessly with widely used vocabularies such as Schema.org, Dublin Core (DCT), and SKOS. This design improves compatibility with existing SEPSES vocabularies, enabling future linking to SEPSES entities such as CVEs or ATT&CK techniques.

3. Cross-Source Semantic Linking

The linking layer also preserves native MISP relationships such as:

misp:Event misp:hasIndicator misp:Attribute .

This enables MISP events and attributes to connect with other models (e.g., CVE or ATT&CK) through identifier-based rules. While the current dataset does not contain automatic CVE or ATT&CK references in the JSON source, the ontology and mapping schemas support these relationships for future data updates.

4. Final Integration

The final KG (full-kg.ttl) is produced by combining:

merged.ttl + linking.ttl

This results in a consistent, semantically enriched graph containing all entities, classes, and alignment properties from the four cybersecurity domains. The structure also enables future enrichment and linking to the existing SEPSES CSKG once identifiers are available in the input data.

5. Reproducibility

All linking files and integration scripts are included in the project’s GitHub repository.

4 Implementation

4.1 Tools and Environment

The CSKG expansion was implemented using the following tools and environment:

- Operating System:
- Programming Environment: Local machine

- RML Engine: RMLMapper.jar (Java-based RML processor)
- Knowledge Graph Storage: Virtuoso Open-Source Edition
- RDF Processing: Apache Jena, terminal-based RDF utilities
- Version Control: GitHub repository https://github.com/Aria1516/Kelompok-1_CSKG
- Ontology Files: Stored under /ontology/
- RML Mapping Files: Stored under /mappings/
- Generated RDF: Stored in /outputs/

4.2 RML Mappings

The RML mappings created in Section 3.3 were executed using RMLMapper.jar, transforming JSON sources into RDF aligned with the four ontologies. Each mapping file was executed using the following command pattern:

```
java -jar rmlmapper.jar -m <mapping-file>.ttl -o <output-file>.ttl
```

Each execution produced one RDF Turtle file:

- attack_output.ttl : ATT&CK techniques and groups
- maec_output.ttl : malware actions and API call structures
- mispf_output.ttl : feed metadata and event structures
- nist_output.ttl : NIST SP800-53 control catalog

These files formed the basis of the integrated CSKG.

4.3 KG Construction Output

After RDF generation, all outputs were merged to produce a single unified KG. The merging process was performed manually and programmatically using Jena tools.

The result was stored in:

- merged.ttl : union of all four RDF outputs
- linking.ttl : vocabulary alignment and inter-model linking
- full-kg.ttl : final integrated CSKG

The final KG (full-kg.ttl) contains:

- All ontology classes and properties
- All instances generated from JSON input sources
- Vocabulary alignments such as:

```
mispf:feedName rdfs:subPropertyOf schema:name .
mispf:feedSlug rdfs:subPropertyOf dct:identifier .
mispf:tagSlug rdfs:subPropertyOf skos:notation .
```

This file represents the complete and queryable CSKG.

4.4 Linking to SEPSES

Although the input datasets did not contain explicit CVE references or ATT&CK technique identifiers linking directly to SEPSES entities, the linking layer was prepared to support SEPSES integration via:

1. Identifier-Based Linking (Ready for Extension)

Examples of intended links:

```
misp:attribute-123 sepses:hasCVE cve:CVE-2021-12345 .
maec:action-456 attack:mapsToTechnique attack:T1041 .
```

2. Ontology Alignment

The linking file includes subproperty alignments to enable future SEPSES compatibility:

```
mispf:feedName rdfs:subPropertyOf schema:name .
mispf:feedSlug rdfs:subPropertyOf dct:identifier .
```

These alignments ensure that SEPSES-based tools or SPARQL queries can consume this KG without modification.

4.5 Deployment

The final knowledge graph (full-kg.ttl) was deployed into Virtuoso as a named graph. The steps were:

1. Access Virtuoso Conductor

Log into the Virtuoso web UI.

2. Upload RDF File

Using:

```
/DAV/home
```

or via:

```
DB.DBA.TTLP_MT(file_to_string_output('full-kg.ttl'), '', 'http://w3id
.org/sepses/cskg-extended');
```

3. Expose SPARQL Endpoint

The KG becomes accessible through the Virtuoso SPARQL interface using its graph IRI.

5 Use-Cases Applications

This section presents three representative use cases that demonstrate the functionality, interoperability, and practical relevance of the constructed Cybersecurity Knowledge Graph (CSKG). The use cases highlight (i) cross-model threat tracing, (ii) gap analysis for quality assurance, and (iii) quantitative assessment of linking and entity distribution. All SPARQL queries and outputs referenced originate from the dataset and examples shown in the attached documentation.

5.1 Use Case 1: End-to-End Threat Traceability Across Heterogeneous Cyber Sources

A central objective of the CSKG is to enable end-to-end traceability of cybersecurity information across heterogeneous data models. This use case demonstrates how a threat observed in a MISP feed can be traced through its associated events and attributes, enriched with MAEC malware actions, aligned with MITRE ATT&CK techniques, and ultimately mapped to NIST Privacy Controls.

A MISP feed produces events, each of which contains one or more indicators (attributes). Attributes may be linked to MAEC malware actions or ATT&CK techniques. MAEC actions may further specify privacy-related impacts that correspond to NIST controls. Listing 1.10 retrieves this full chain of relationships:

Listing 1.10: SPARQL Query for End-to-End Threat Traceability

```
PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
PREFIX mispf: <http://w3id.org/sepses/vocab/ref/misp/feed/>
PREFIX maec: <http://w3id.org/sepses/vocab/ref/maec#>
PREFIX attack: <http://w3id.org/sepses/vocab/ref/attack#>
PREFIX nistp: <http://w3id.org/sepses/vocab/ref/nist-privacy#>
SELECT DISTINCT ?feed ?event ?attribute ?maec ?technique ?control
WHERE {
  ?feed a misp:MISPFeed .
  ?feed misp:producesEvent ?event .
  ?event misp:hasIndicator ?attribute .
  OPTIONAL { ?attribute misp:linkedMAEC ?maec . }
  OPTIONAL { ?attribute misp:linkedAttack ?technique . }
  OPTIONAL { ?maec maec:impactsPrivacyControl ?control . }
}
LIMIT 200
```

The expected output forms the chain:

Feed → *Event* → *Attribute* → *MAEC Action* → *ATT&CK Technique* → *NIST Privacy Control*.

As illustrated in Fig. 3, this confirms that the CSKG successfully integrates isolated cybersecurity standards and enables holistic threat tracing.

5.2 Use Case 2: Gap Analysis for Incomplete or Missing Links in the Knowledge Graph

Gap analysis evaluates the completeness and consistency of the CSKG. This use case employs SPARQL queries to detect missing links or unmapped entities, allowing developers to identify weaknesses in the extraction pipeline or ontology alignment. Three gap categories are examined:

1. Events Without Attributes

As shown in Listing 1.11, this query identifies events that do not contain any indicators.

Listing 1.11: SPARQL Query for Events Without Attributes

```

PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
SELECT ?event
WHERE {
  ?event a misp:Event .
  FILTER NOT EXISTS { ?event misp:hasIndicator ?attr }
}

```

Figure 4 shows the events that have no linked attributes.

2. Attributes Without MAEC or ATT&CK Mappings

As shown in Listing 1.12, this query identifies attributes lacking semantic enrichment.

Listing 1.12: SPARQL Query for Attributes Missing MAEC or ATT&CK Links

```

PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
SELECT ?attr
WHERE {
  ?attr a misp:Attribute .
  FILTER NOT EXISTS { ?attr misp:linkedMAEC ?m }
  FILTER NOT EXISTS { ?attr misp:linkedAttack ?t }
}

```

As shown in Figure 5, several attributes lack enrichment links.

3. MAEC Actions Without Corresponding ATT&CK Techniques

MAEC malware actions that have not been mapped to ATT&CK techniques are detected using the following query in Listing 1.13.

Listing 1.13: SPARQL Query for MAEC Actions Without ATT&CK Mappings

```

PREFIX maec: <http://w3id.org/sepses/vocab/ref/maec#>
SELECT ?maec
WHERE {
  ?maec a maec:MalwareAction .
  FILTER NOT EXISTS { ?maec maec:mapsToAttack ?t }
}

```

Figure 6 highlights MAEC actions missing ATT&CK mappings.

These results reveal events lacking indicators, attributes without enrichment links, and malware actions not aligned with standardized attack techniques. These findings guide iterative improvements in mapping rules, RML pipelines, and ontology design.

5.3 Use Case 3: Quantitative Evaluation of Linking Density and Entity Distribution

This use case quantifies graph density and cross-model connectivity. This analysis helps evaluate how well the newly constructed CSKG integrates entities from MISP, MAEC, ATT&CK, and NIST, and identifies areas requiring additional enrichment. Three metrics are evaluated:

1. Event Count per Feed

Listing 1.14: SPARQL Query for Event Count per Feed

```
PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
SELECT ?feed (COUNT(?event) AS ?numEvents)
WHERE {
  ?feed a misp:MISPFee .
  ?feed misp:producesEvent ?event .
}
GROUP BY ?feed
```

This query yields the number of events produced by each feed, as shown in Listing 1.14. Disproportionate event counts may reflect highly active feeds or potential extraction inconsistencies.

As shown in Figure 7, each MISP feed generates an equal number of events.

2. MAEC Actions per ATT&CK Technique

Listing 1.15: SPARQL Query for MAEC Actions per ATT&CK Technique

```
PREFIX maec: <http://w3id.org/sepses/vocab/ref/maec#>
PREFIX attack: <http://w3id.org/sepses/vocab/ref/attack#>
SELECT ?technique (COUNT(?maec) AS ?numActions)
WHERE {
  ?maec maec:mapsToAttack ?technique .
}
GROUP BY ?technique
ORDER BY DESC(?numActions)
```

This query measures how many malware actions correspond to each ATT&CK technique, as shown in Listing 1.15. A higher number indicates that the technique is central to observed malware behavior.

Figure 8 lists the techniques with the highest number of mapped MAEC actions.

3. ATT&CK Techniques per NIST Privacy Control

Listing 1.16: SPARQL Query for ATT&CK Techniques per NIST Privacy Control

```
PREFIX attack: <http://w3id.org/sepses/vocab/ref/attack#>
PREFIX nistp: <http://w3id.org/ns/sepses/vocab/ref/nist-privacy#>
SELECT ?control (COUNT(?technique) AS ?numTechniques)
WHERE {
  ?technique attack:hasPrivacyImpact ?control .
}
GROUP BY ?control
ORDER BY DESC(?numTechniques)
```

As shown in Figure 9, NIST controls C06 and C01 have the highest number of mapped techniques.

The results show which privacy controls are most frequently impacted by adversarial behavior. Techniques linked to multiple controls indicate broader privacy implications and may be prioritized for risk mitigation.

6 Statistics, Evaluation, and Discussion

This section presents the statistical summary, structural evaluation, and quality assessment of the constructed Cybersecurity Knowledge Graph (CSKG), which integrates four ontologies: MITRE ATT&CK, MAEC, MISP Feed, and NIST Cybersecurity Framework. Evaluation includes triple counts, class and property distribution, SHACL validation, missing values, limitations, and visual representations.

6.1 Summary Statistics of the Constructed KG

Table 1 presents the number of triples produced by each ontology during the CSKG construction process. The MITRE ATT&CK ontology contributes the largest share, with 7,575 triples, reflecting its extensive and well-structured representation of tactics, techniques, software, and adversary behaviors. MISP Feed adds 1,656 triples, mainly consisting of threat intelligence metadata and feed configuration details. NIST Controls contributes 361 triples, enriching the graph with governance and mitigation-related information. MAEC, being a compact and highly granular ontology focused on malware actions and API calls, contributes only 99 triples. The integrated knowledge graph contains a total of 14,926 triples, demonstrating successful merging and linking across all ontologies.

Table 1: Triple Counts per Source and Final Integrated Graph

Source Ontology	Triples	Notes
MITRE ATT&CK	7,575	Largest contributor; rich behavior relationships
MAEC	99	Small ontology; high granularity, low volume
MISP Feed	1,656	Metadata-rich, moderate size
NIST Controls	361	High data property density
Full Integrated KG	14,926	Includes linking triples across ontologies

Figure 2 visualizes the percentage distribution of triples across the four ontologies. MITRE ATT&CK accounts for 50.7% of all triples, confirming its role as the central behavioral backbone in the integrated KG. MISP Feed provides 11.1% of triples, supplying important threat intelligence context. NIST contributes 2.4%, representing control and mitigation concepts that complement the behavioral layer. MAEC contributes 0.6%, consistent with its intentionally narrow focus on low-level malware behaviors. Overall, the distribution highlights the multi-domain nature of the constructed KG, with ATT&CK dominating the behavioral layer while MISP, NIST, and MAEC contribute supporting semantic information.

6.2 Class and Property Distribution

Table 2 summarizes the structural characteristics of each ontology used in the construction of the Cybersecurity Knowledge Graph. The MITRE ATT&CK ontology contains the largest number of classes and object properties, reflecting its comprehensive modeling of adversary behaviors, techniques, tactics, and related entities. MAEC, on the

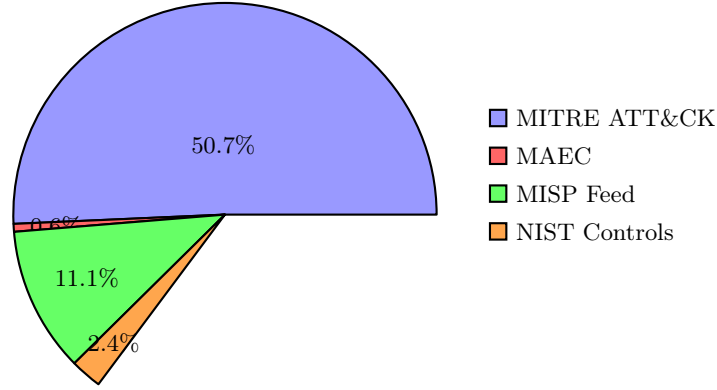


Fig. 2: Percentage Contribution of Triples Across Ontology Sources

other hand, has a small class hierarchy but a relatively high number of data properties due to its focus on detailed malware actions and API call parameters.

MISP Feed includes extensive annotation properties, which aligns with its purpose of representing threat intelligence metadata and feed configuration information. NIST Controls provides a moderate number of classes and properties, mainly representing governance, control activities, and mitigation concepts. Overall, the table highlights the varying complexity and scope of each ontology, demonstrating their complementary roles in enriching different semantic layers of the integrated knowledge graph.

Table 2: Structural Metrics for Each Ontology

Ontology	Classes	Object Properties	Data Properties	Annotation Properties
MITRE ATT&CK	14	15	10	17
MAEC	3	2	13	23
MISP Feed	4	2	11	72
NIST Controls	7	5	17	40

6.3 Coverage Analysis per Source

The coverage of each ontology within the integrated KG can be observed through triple distribution:

- ATT&CK (highest coverage): Strong attack-behavior layer (Tactics, Techniques, Software, Threat Groups).
- MAEC: Provides execution-level behavior that augments ATT&CK techniques using low-level API actions.
- MISP: Adds threat intelligence provenance including feed information, object identifiers, and indicators.

- NIST: Connects operational and governance controls to ATT&CK techniques (mitigation mapping).

The KG achieves both horizontal coverage (multiple cybersecurity domains) and vertical depth (from governance to behavior to technical indicators).

6.4 SHACL Validation Report

All ontologies and the final merged knowledge graph were validated using pySHACL against custom SHACL shape constraints for each ontology (ATT&CK, MAEC, MISP, NIST). Validation was run individually per ontology output and on the full integrated KG.

Table 3: SHACL Validation Results

Validation Target	SHACL Result	Violations
ATT&CK Output	Conforms	0
MAEC Output	Conforms	0
MISP Output	Conforms	0
NIST Output	Conforms	0
Full KG (Merged)	Conforms	0

The resulting KG is structurally sound, complete, and semantically consistent with respect to the defined SHACL constraints.

6.5 Evaluation

Because SHACL validation returned zero violations, there are no missing values in required properties across all four ontologies. Despite full SHACL conformance, several limitations remain:

1. Coverage imbalance: ATT&CK contributes disproportionately, meaning the KG is behavior centric.
2. Small MAEC footprint: Although valuable for behavioral enrichment, MAEC contributes very few triples due to its compact ontology structure.
3. Dependency on source completeness: MISP feed and NIST control implementation levels depend heavily on source data quality.
4. No temporal reasoning: Time-based relationships (e.g., incident timelines) are not modeled.
5. Limited inference: Only structural validation was tested; reasoning-based validation was not performed.

7 Conclusion

This work extends the SEPSES Cybersecurity Knowledge Graph (CSKG) by integrating four major cybersecurity data models MITRE ATT&CK, MAEC, MISP, and

NIST SP 800-53 into a unified semantic framework. Using RML mappings, JSON-based threat intelligence sources were transformed into ontology-aligned RDF graphs, merged, and enriched through vocabulary alignment and cross-model linking. The resulting integrated KG (full-kg.ttl) supports interoperable reasoning across heterogeneous cybersecurity standards.

The implementation demonstrates that a multi-ontology integration approach enables end-to-end threat traceability, automated gap identification, and quantitative assessment of entity relationships. These capabilities highlight the practical value of a unified CSKG for improving incident analysis, threat correlation, and security control mapping. Although the source datasets did not yet include CVE or ATT&CK identifiers required for full SEPSES cross-linking, the linking schema and ontology design have been prepared to support seamless extension when such identifiers become available.

Overall, the expanded CSKG provides a scalable foundation for deeper cybersecurity analytics and opens opportunities for future work such as automated CVE-ATT&CK linkage, real-time feed ingestion, and integration with existing SEPSES knowledge services.

SPARQL Execution

Query: `PREFIX msp: <http://w3id.org/semantics/vocab/misp/>
 PREFIX attack: <http://w3id.org/semantics/vocab/attack/>
 PREFIX msp: <http://w3id.org/semantics/vocab/misp/>
 PREFIX msp: <http://w3id.org/semantics/vocab/misp/>

 SELECT DISTINCT ?feed ?event ?attribute ?misp ?technique ?control
 WHERE {
 ?feed a msp:Feed .
 ?event msp:proceedsFrom ?feed .
 ?attribute msp:hasIndicator ?event .
 ?misp msp:hasIndicator ?attribute .

 OPTIONAL { ?attribute msp:hasMAEC ?misp . }
 OPTIONAL { ?attribute msp:hasATTCK ?technique . }
 OPTIONAL { ?misp msp:hasNIST ?control . }
 }
 LIMIT 200`

SPARQL | HTMLS table

feed	event	attribute	misp	technique	control
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_21	http://w3id.org/semantics/vocab/misp/attr_101	http://w3id.org/semantics/vocab/misp/action_82	http://w3id.org/semantics/vocab/misp/attack_771029	http://w3id.org/semantics/vocab/misp/control_4212
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_21	http://w3id.org/semantics/vocab/misp/attr_102	http://w3id.org/semantics/vocab/misp/action_111	http://w3id.org/semantics/vocab/misp/attack_771022	http://w3id.org/semantics/vocab/misp/control_4216
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_21	http://w3id.org/semantics/vocab/misp/attr_103	http://w3id.org/semantics/vocab/misp/action_189	http://w3id.org/semantics/vocab/misp/attack_771029	http://w3id.org/semantics/vocab/misp/control_4207
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_23	http://w3id.org/semantics/vocab/misp/attr_143	http://w3id.org/semantics/vocab/misp/action_199	http://w3id.org/semantics/vocab/misp/attack_771021	http://w3id.org/semantics/vocab/misp/control_4205
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_3	http://w3id.org/semantics/vocab/misp/attr_113	http://w3id.org/semantics/vocab/misp/action_131	http://w3id.org/semantics/vocab/misp/attack_771028	http://w3id.org/semantics/vocab/misp/control_4215
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_3	http://w3id.org/semantics/vocab/misp/attr_113	http://w3id.org/semantics/vocab/misp/action_147	http://w3id.org/semantics/vocab/misp/attack_771026	http://w3id.org/semantics/vocab/misp/control_4212
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_29	http://w3id.org/semantics/vocab/misp/attr_143	http://w3id.org/semantics/vocab/misp/action_199	http://w3id.org/semantics/vocab/misp/attack_771021	http://w3id.org/semantics/vocab/misp/control_4205
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_29	http://w3id.org/semantics/vocab/misp/attr_143	http://w3id.org/semantics/vocab/misp/action_149	http://w3id.org/semantics/vocab/misp/attack_771011	http://w3id.org/semantics/vocab/misp/control_4207
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_30	http://w3id.org/semantics/vocab/misp/attr_148	http://w3id.org/semantics/vocab/misp/action_153	http://w3id.org/semantics/vocab/misp/attack_771024	http://w3id.org/semantics/vocab/misp/control_4209
http://w3id.org/semantics/vocab/misp/feed_3	http://w3id.org/semantics/vocab/misp/event_30	http://w3id.org/semantics/vocab/misp/attr_148	http://w3id.org/semantics/vocab/misp/action_79	http://w3id.org/semantics/vocab/misp/attack_771024	http://w3id.org/semantics/vocab/misp/control_4207

Fig. 3: End-to-end threat traceability from MISP feeds to MAEC actions, ATT&CK techniques, and NIST controls.

SPARQL Execution

Query: `PREFIX msp: <http://w3id.org/semantics/vocab/misp/>
 PREFIX attack: <http://w3id.org/semantics/vocab/attack/>
 PREFIX msp: <http://w3id.org/semantics/vocab/misp/>
 PREFIX msp: <http://w3id.org/semantics/vocab/misp/>

 SELECT DISTINCT ?feed ?event ?attribute ?misp ?technique ?control
 WHERE {
 ?feed a msp:Feed .
 ?event msp:proceedsFrom ?feed .
 ?attribute msp:hasIndicator ?event .
 ?misp msp:hasIndicator ?attribute .

 OPTIONAL { ?attribute msp:hasMAEC ?misp . }
 OPTIONAL { ?attribute msp:hasATTCK ?technique . }
 OPTIONAL { ?misp msp:hasNIST ?control . }
 }
 LIMIT 200`

SPARQL | HTMLS table

event
http://w3id.org/semantics/vocab/misp/event_1
http://w3id.org/semantics/vocab/misp/event_2
http://w3id.org/semantics/vocab/misp/event_3
http://w3id.org/semantics/vocab/misp/event_4
http://w3id.org/semantics/vocab/misp/event_5
http://w3id.org/semantics/vocab/misp/event_6
http://w3id.org/semantics/vocab/misp/event_7
http://w3id.org/semantics/vocab/misp/event_8
http://w3id.org/semantics/vocab/misp/event_9
http://w3id.org/semantics/vocab/misp/event_10

Fig. 4: Gap Analysis: MISP events that do not contain any associated attributes.

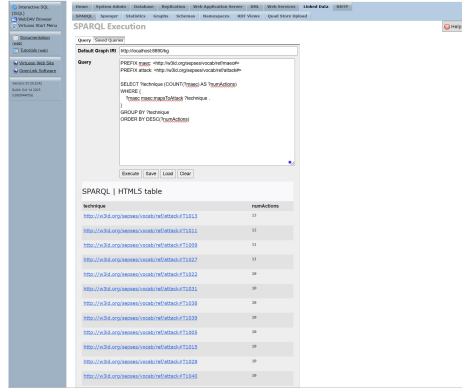


Fig. 8: Distribution of MAEC malware actions mapped to each ATT&CK technique.

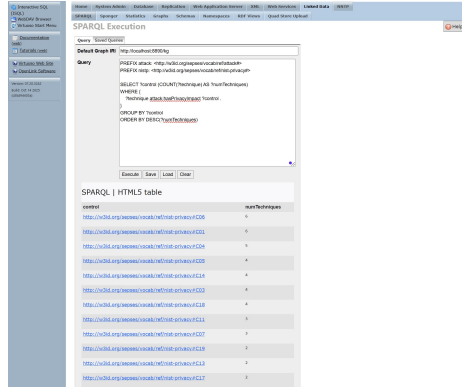


Fig. 9: Mapping density: number of ATT&CK techniques associated with each NIST privacy control.

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